

Candidate Full Result: The Random-Semigroup SLLN Holds on Every Banach Space

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Status. **Candidate full solution, likely valid, needs human review.** This packet proves Conjecture 2.6 of Dzhenzher–Sakbaev [1]: the strong-operator strong law for bounded i.i.d. random semigroup generators does not require uniform smoothness of the Banach space.

Source Conjecture

The source paper proves the result in strong operator topology when $\mathcal{X}$ is uniformly smooth, and in weak operator topology for arbitrary Banach spaces. On page 4, after noting the ℓ_1 consequence, it states Conjecture 2.6:

$$\varepsilon \rightarrow 0 \quad N = 1/n \rightarrow \infty$$

where the first equality is up to a set of zero probability.

Let us prove the «only if» part. Take a separable subset $\mathcal{X}_0 \subset \mathcal{X}$ in which ξ lies almost surely. For each $n \in \mathbb{N}$ cover $\mathcal{X}_0$ with a countable family of balls $B_{n,k}$ of radius $\frac{1}{n}$. For each n modify these $B_{n,k}$ so that $B_{n,k+1}$ is obtained by deleting $B_{n,k}$. They become a disjoint cover of $\mathcal{X}_0$. Construct countably valued functions ξ_n by setting on $\xi^{-1}(B_{n,k})$ some points from $B_{n,k}$, and on the remaining set of zero probability some fixed point. Then by [33, Corollary II.1.3] $\xi_n \xrightarrow[n \rightarrow \infty]{u.a.s.} \xi$ implies Bochner measurability of ξ . $\square$

Random elements $\{\xi_n: \Omega \rightarrow \mathcal{X}_n\}$ are **independent** if for any $n \in \mathbb{N}$ and any open $O_k \subset \mathcal{X}_k$ we have

$$P\{\xi_1 \in O_1, \dots, \xi_n \in O_n\} = P\{\xi_1 \in O_1\} \dots P\{\xi_n \in O_n\}.$$

Note that one may take $O_k = \mathcal{X}_k$ and so make some events be the whole Ω .

Semigroups. For a bounded random operator $A: \Omega \rightarrow \mathcal{L}(\mathcal{X})$ denote by e^{At} the random (uniformly continuous) semigroup with A as the infinitesimal generator.

Theorem 2.4 (SLLN in SOT; proved in §4). *Let $\mathcal{X}$ be a uniformly smooth Banach space. Let $A: \Omega \rightarrow \mathcal{L}(\mathcal{X})$ be a generator of a semigroup e^{At} such that $\|A\|$ is bounded. Let $A, A_1, A_2, \dots$ be a sequence of i.i.d. random generators. Then $e^{A_1 t/n} \dots e^{A_n t/n}$ converges a.s. in SOT of $\|\cdot\|_{\mathcal{X}}$ to $e^{\mathbb{E}A}$ uniformly for $t > 0$ in any segment; that is, for any $x \in \mathcal{X}$ and $T > 0$ almost surely*

$$\lim_{n \rightarrow \infty} \sup_{t \in [0, T]} \|(e^{A_1 t/n} \dots e^{A_n t/n} - e^{\mathbb{E}A})x\| = 0.$$

There is an example [5, Example 5] when the LLN fails even for a Hilbert space $\mathcal{X}$ in the case of a finitely additive measure (but not σ -additive). In that case, Bochner measurability is replaced with weak measurability, and expected value becomes the Pettis integral.

It is interesting to notice that in the usual Strong Law of Large Numbers for random variables it is necessary to demand only that all variables are uniformly bounded by some constant, independent and have the same expected value. So in the classical case the requirement of the random variables being i.i.d. is redundant. However, it is not clear if this requirement is redundant in the case of random semigroups of operators.

Theorem 2.5 (SLLN in WOT; proved in §5). *Let $\mathcal{X}$ be a Banach space. Then under the assumptions of Theorem 2.4, $e^{A_1 t/n} \dots e^{A_n t/n}$ converges a.s. in WOT to $e^{\mathbb{E}A}$ uniformly for $t > 0$ in any segment; that is, for any $x \in \mathcal{X}$, $f \in \mathcal{X}^*$ and $T > 0$ almost surely*

$$\lim_{n \rightarrow \infty} \sup_{t \in [0, T]} |\langle f, (e^{A_1 t/n} \dots e^{A_n t/n} - e^{\mathbb{E}A})x \rangle| = 0.$$

Thus by the Schur's Theorem we immediately obtain that for $\mathcal{X} = \ell_1$ the SLLN in SOT of $\|\cdot\|_{\ell_1}$ holds. Note that ℓ_1 is not uniformly smooth. This leads us to the following conjecture.

Conjecture 2.6. *Let $\mathcal{X}$ be a Banach space. Then under the assumptions of Theorem 2.4, $e^{A_1 t/n} \dots e^{A_n t/n}$ converges a.s. in SOT of $\|\cdot\|_{\mathcal{X}}$ to $e^{\mathbb{E}A}$ uniformly for $t > 0$ in any segment.*

Transcribed, the conjecture says:

Let $\mathcal{X}$ be a Banach space. Then under the assumptions of Theorem 2.4, $e^{A_1 t/n} \dots e^{A_n t/n}$ converges a.s. in SOT of $\|\cdot\|_{\mathcal{X}}$ to $e^{\mathbb{E}A}$ uniformly for $t > 0$ in any segment.

Here $A, A_1, A_2, \dots$ are i.i.d. bounded random elements of $\mathcal{L}(\mathcal{X})$, and $\mathbb{E}A$ is the Bochner expectation.

Proof Intuition

The source proof uses uniform smoothness to obtain summable martingale moment estimates for the random product. For bounded generators there is a more direct pathwise route. First replace each exponential $e^{A_j t/n}$ by the Euler factor $I + (t/n)A_j$; the accumulated replacement error is $O(1/n)$, uniformly for t in a compact interval. The Euler product is then compared with the deterministic Euler product for $\bar{A} = \mathbb{E}A$.

For a fixed t , the accumulated error solves a linear recursion driven by mean-zero increments

$$\frac{t}{n}(A_j - \bar{A})y_{n,j},$$

where $y_{n,j}$ lies on the deterministic Euler orbit of x . That orbit is contained in a compact curve. A finite partition of the curve reduces the noise estimate to finitely many ordinary Banach-valued strong laws of large numbers, applied on intervals of indices. A discrete summation-by-parts estimate then converts small partial sums of the noise into small product error. Finally, a uniform Lipschitz bound in t upgrades convergence on rational t 's to uniform convergence on every compact interval.

Statement

Theorem 1. *Let $\mathcal{X}$ be a Banach space. Let $A, A_1, A_2, \dots$ be i.i.d. Bochner-measurable random elements of $\mathcal{L}(\mathcal{X})$, and assume that $\|A\| \leq \rho$ almost surely for some $\rho < \infty$. Put $\bar{A} = \mathbb{E}A$. Then for every $x \in \mathcal{X}$ and every $T > 0$,*

$$\lim_{n \rightarrow \infty} \sup_{0 \leq t \leq T} \left\| \left(e^{A_1 t/n} \dots e^{A_n t/n} - e^{\bar{A}t} \right) x \right\| = 0$$

almost surely.

Proof

We use the following elementary form of the Banach-valued strong law. It is a standard consequence of the usual SLLN for Bochner-integrable i.i.d. Banach-valued random variables; boundedness is more than enough for the application below.

Lemma 1 (Interval SLLN). *Let $Z_1, Z_2, \dots$ be i.i.d. bounded random variables with values in a Banach space E , and assume $\mathbb{E}Z_1 = 0$. Then, almost surely,*

$$\sup_{0 \leq a \leq b \leq n} \left\| \frac{1}{n} \sum_{j=a+1}^b Z_j \right\|_E \rightarrow 0.$$

Proof. The ordinary Banach-valued SLLN gives $m^{-1} \sum_{j=1}^m Z_j \rightarrow 0$ almost surely. Hence

$$\max_{0 \leq m \leq n} \left\| \frac{1}{n} \sum_{j=1}^m Z_j \right\|_E \rightarrow 0$$

almost surely: finitely many initial partial sums are divided by n , while the remaining ones are controlled by the SLLN. The interval estimate follows by subtracting two partial sums. $\square$

Lemma 2 (Compact-curve noise estimate). *Let $B_j = A_j - \bar{A}$. Fix $x \in \mathcal{X}$ and $t \geq 0$. Let $h_n = t/n$ and*

$$y_{n,k} = (I + h_n \bar{A})^k x, \quad 0 \leq k \leq n.$$

Then, almost surely,

$$\max_{0 \leq m \leq n} \left\| \frac{1}{n} \sum_{k=1}^m B_{n-k+1} y_{n,k-1} \right\| \rightarrow 0.$$

Proof. The points $y_{n,k}$ converge uniformly, as sampled Euler points, to the compact curve $s \mapsto e^{s\bar{A}}x$, $0 \leq s \leq t$. Given $\varepsilon > 0$, choose a partition

$$0 = s_0 < s_1 < \cdots < s_M = t$$

so fine that both $e^{s\bar{A}}x$ and the Euler samples vary by at most ε inside each subinterval, for all sufficiently large n . For k with $(k-1)t/n \in [s_{r-1}, s_r)$, replace $y_{n,k-1}$ by the fixed vector $z_r = e^{s_{r-1}\bar{A}}x$. Since $\|B_j\| \leq 2\rho$, the total error in the normalized sums is at most $2\rho\varepsilon$.

It remains to handle finitely many fixed vectors z_r . For each r , the variables $B_j z_r$ are i.i.d., bounded, and have mean zero. Lemma 1 gives

$$\sup_{0 \leq a \leq b \leq n} \left\| \frac{1}{n} \sum_{j=a+1}^b B_j z_r \right\| \rightarrow 0$$

almost surely. The sums appearing after the replacement are finite sums of such interval sums, because $k \mapsto n - k + 1$ reverses intervals. Intersect the finitely many probability-one events and let $\varepsilon \downarrow 0$. $\square$

Lemma 3 (Euler products). *For each fixed $t \geq 0$,*

$$\left(I + \frac{t}{n} A_1\right) \cdots \left(I + \frac{t}{n} A_n\right) x - \left(I + \frac{t}{n} \bar{A}\right)^n x \rightarrow 0$$

almost surely.

Proof. Write $h = t/n$, $C_j = I + hA_j$, and $\bar{C} = I + h\bar{A}$. To match the source product order, define

$$X_{n,0} = x, \quad X_{n,k} = C_{n-k+1} X_{n,k-1}, \quad 1 \leq k \leq n,$$

so $X_{n,n} = C_1 \cdots C_n x$. Also put $y_{n,k} = \bar{C}^k x$ and $D_{n,k} = X_{n,k} - y_{n,k}$. Then

$$D_{n,k} = C_{n-k+1} D_{n,k-1} + h(A_{n-k+1} - \bar{A}) y_{n,k-1}.$$

Set

$$\eta_{n,k} = h(A_{n-k+1} - \bar{A}) y_{n,k-1}, \quad S_{n,m} = \sum_{k=1}^m \eta_{n,k}.$$

By Lemma 2, $\max_{m \leq n} \|S_{n,m}\| \rightarrow 0$ almost surely.

Solving the linear recursion gives

$$D_{n,m} = \sum_{k=1}^m R_{m,k} \eta_{n,k},$$

where $R_{m,m} = I$ and

$$R_{m,k} = C_{n-m+1} \cdots C_{n-k} \quad (k < m).$$

The exact indexing is unimportant; only the following bounds matter:

$$\|R_{m,k}\| \leq (1 + \rho h)^{m-k} \leq e^{\rho t}$$

and

$$\|R_{m,k} - R_{m,k+1}\| \leq \rho h e^{\rho t}.$$

By Abel summation,

$$D_{n,m} = R_{m,m} S_{n,m} + \sum_{k=1}^{m-1} (R_{m,k} - R_{m,k+1}) S_{n,k}.$$

Consequently

$$\max_{m \leq n} \|D_{n,m}\| \leq (1 + \rho t e^{\rho t}) \max_{m \leq n} \|S_{n,m}\|,$$

which tends to zero almost surely. Taking $m = n$ proves the lemma. $\square$

Proof of Theorem 1. First fix $t \in [0, T]$. Lemma 3 gives almost sure convergence of the random Euler product to $(I + (t/n)\bar{A})^n x$, and the latter converges in norm to $e^{t\bar{A}}x$.

It remains to pass from Euler factors to exponentials and then make the convergence uniform in t . Since $\|A_j\| \leq \rho$,

$$\|e^{hA_j} - (I + hA_j)\| \leq \frac{\rho^2 h^2}{2} e^{\rho h}.$$

Telescoping the product and using $\|e^{hA_j}\| \leq e^{\rho h}$, $\|I + hA_j\| \leq 1 + \rho h$, gives

$$\sup_{0 \leq t \leq T} \left\| \left(e^{A_1 t/n} \dots e^{A_n t/n} - \left(I + \frac{t}{n} A_1 \right) \dots \left(I + \frac{t}{n} A_n \right) \right) x \right\| \leq \frac{C_{\rho,T} \|x\|}{n}.$$

The same deterministic estimate gives

$$\sup_{0 \leq t \leq T} \left\| \left(I + \frac{t}{n} \bar{A} \right)^n x - e^{t\bar{A}} x \right\| \rightarrow 0.$$

Thus the exponential product converges almost surely for every fixed rational $t \in [0, T]$, after intersecting countably many probability-one events.

Finally, the maps

$$t \mapsto e^{A_1 t/n} \dots e^{A_n t/n} x \quad \text{and} \quad t \mapsto e^{t\bar{A}} x$$

are Lipschitz on $[0, T]$ with a common deterministic bound

$$\rho e^{\rho T} \|x\|.$$

Indeed, differentiate the products and sum the n terms, each contributing at most $(\rho/n)e^{\rho T} \|x\|$. Convergence on a dense rational set therefore implies uniform convergence on $[0, T]$. This proves the conjecture. $\square$

Verification Notes

The proof uses only boundedness, Bochner measurability, independence, and identical distribution of the generators. Uniform smoothness is not used. The argument does not settle the different issue mentioned immediately before Theorem 2.5 in the source paper: whether the i.i.d. assumption can be weakened to independent uniformly bounded generators with a common expectation.

A small finite-dimensional smoke script in `code/matrix_smoke.py` checks the theorem numerically for random 2×2 matrix generators. It is only a sanity check; the proof above is analytic.

Novelty Check

The local run indexes were searched for arXiv ‘2507.07658’, the source title, ‘random semigroups’, ‘strong law of large numbers’, ‘SOT’, and related Banach-space keywords. No prior packet or attempt for this conjecture was found. Web searches on 2026-06-26 for the exact title, ‘Conjecture 2.6 random semigroups Banach space’, ‘SLLN in SOT random semigroups Banach space’, and close variants found the source arXiv paper and a 2026 follow-up by the same authors on unbounded generators in uniformly smooth Banach spaces, but no paper resolving the bounded-generator conjecture for arbitrary Banach spaces.

Human Review Recommendation

Review should focus on Lemma 2, especially the interval SLLN reduction for the reversed index order, and on the Abel summation step in Lemma 3. If those two estimates pass, the packet gives a full positive solution of Conjecture 2.6.

References

- [1] S. V. Dzhenezher and V. Zh. Sakbaev, *The Strong Law of Large Numbers for random semigroups on uniformly smooth Banach spaces*, arXiv:2507.07658, 2025.
- [2] M. Ledoux and M. Talagrand, *Probability in Banach Spaces: Isoperimetry and Processes*, Springer, 1991.